



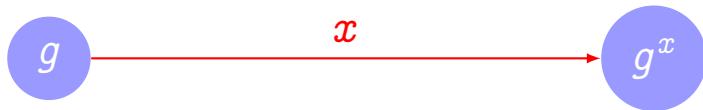
How to solve the supersingular endomorphism ring problem

Luca De Feo

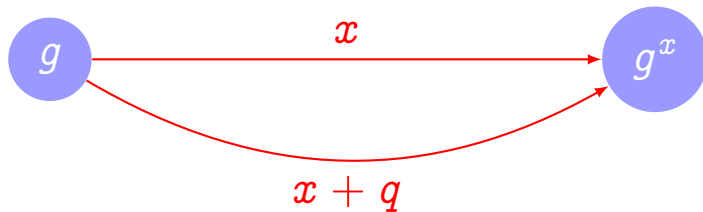
IBM Research Zürich

September 7, 2026, Mathematics of Post-Quantum Cryptography, Raitenhaslach

A group



A group



Several generators

$$g_1 \quad g_2 \quad g_3 \quad g_4 \quad g_5$$

Several generators

$$g_1^7 \quad g_2^{19} \quad g_3^{39} \quad g_4^{67} \quad g_5^0 = 1$$

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$$g_1^{27} \quad g_2^{39} \quad g_3^{51} \quad g_4^{63} \quad g_5^{75} = 1$$

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$$\mathbb{Z}^5 / \Lambda \longrightarrow G$$

The isogeny problem



The isogeny problem



Algebra



Geometry

Isogenies



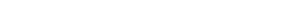
Algebraic morphisms: $\left(\frac{5x^2 + 5x + 4}{x + 1}, y \frac{2x^2 + 4x - 4}{x^2 + 2x + 1} \right)$

Group morphisms: $\varphi(P + Q) = \varphi(P) + \varphi(Q)$

Isogenies



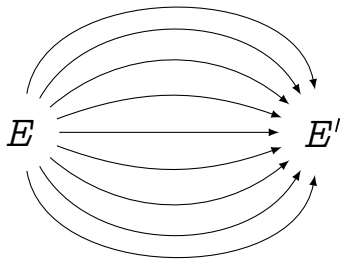
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Group morphisms: $\varphi(P + Q) = \varphi(P) + \varphi(Q)$

Groups of morphisms

“Classes” of isogenies



$$\mathrm{Hom}(E, E')$$

Degree

$$\deg(\varphi) \in \mathbb{N}^+$$

- A measure of “size”;



Degree

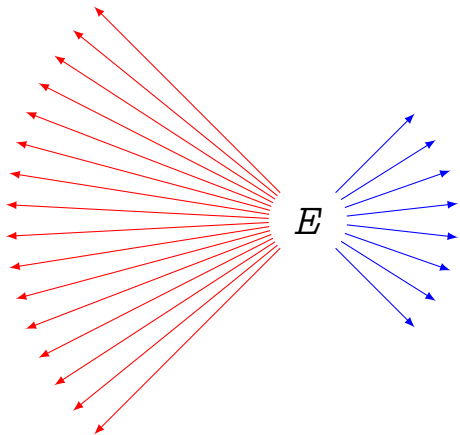
$$\deg(\varphi) \in \mathbb{N}^+$$

- A measure of “size”;
- Multiplicative.

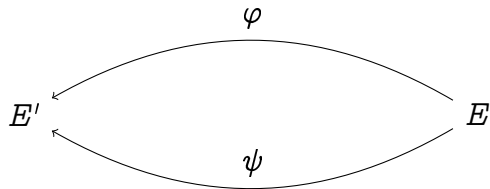


$$\deg(\psi \circ \varphi) = \deg(\psi) \deg(\varphi)$$

The larger the degree, the more isogenies

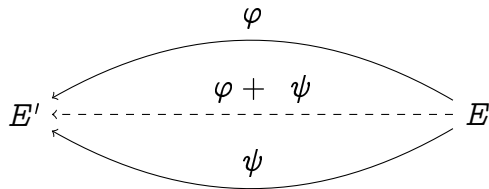


Lattices of isogenies



$$\varphi, \psi \in \text{Hom}(E, E')$$

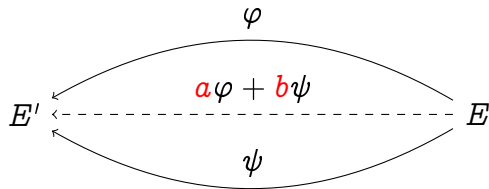
Lattices of isogenies



$$\varphi, \psi \in \text{Hom}(E, E')$$

$$(\varphi + \psi)(P) := \varphi(P) + \psi(P)$$

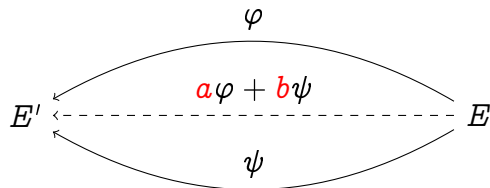
Lattices of isogenies



$$\varphi, \psi \in \text{Hom}(E, E')$$

$$(a\varphi + b\psi)(P) := [a]\varphi(P) + [b]\psi(P)$$

Lattices of isogenies

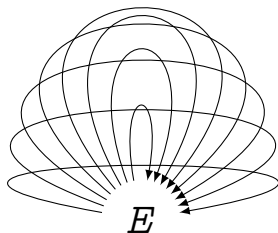


$$\varphi, \psi \in \text{Hom}(E, E')$$

$$(a\varphi + b\psi)(P) := [a]\varphi(P) + [b]\psi(P)$$

- $\deg(a\varphi) = a^2 \deg(\varphi)$,
- $\left| \deg(\varphi + \psi) - \deg(\varphi) - \deg(\psi) \right| \leq 2\sqrt{\deg(\varphi) \deg(\psi)}$,
- $\deg$ is a positive definite quadratic form.

Endomorphisms



$$(\varphi + \psi)(P) := \varphi(P) + \psi(P)$$

$$(\varphi \circ \psi)(P) := \varphi(\psi(P))$$

$$\varphi \circ (\psi + \chi) = \varphi \circ \psi + \varphi \circ \chi$$

Endomorphism rings

$$\mathbb{Q}(\sqrt{-p})$$

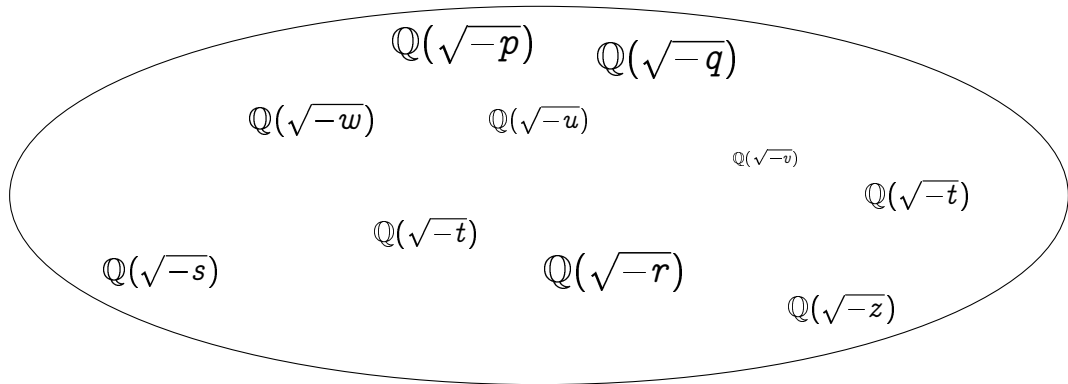
Quadratic imaginary field

Endomorphism rings

$$\mathbb{Q}(\sqrt{-p}) \quad \mathbb{Q}(\sqrt{-q})$$

$$i^2 = -p \quad j^2 = -q \quad ij = -ji$$

Endomorphism rings



Quaternion algebra

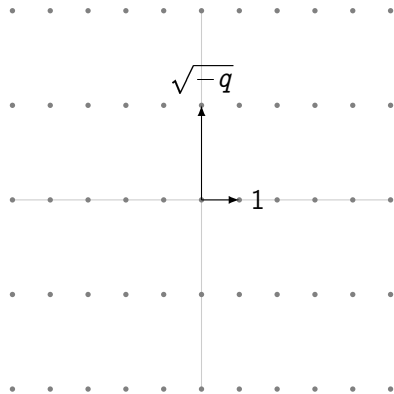
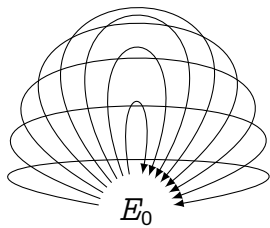
$$i^2 = -p \quad j^2 = -q \quad ij = -ji$$

Orders

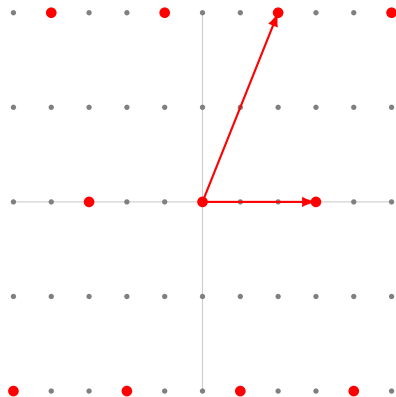
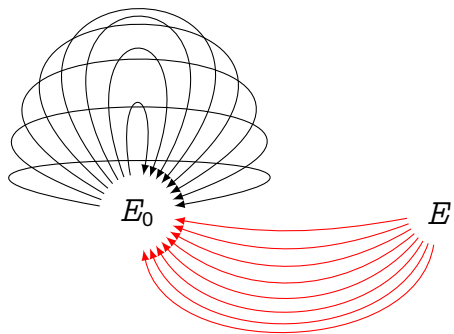
$$\frac{1}{2} \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ i \\ j \\ ij \end{pmatrix}$$

Order type:

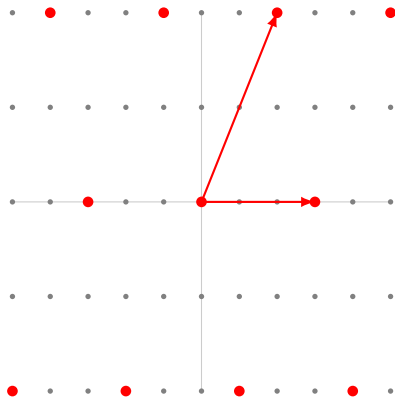
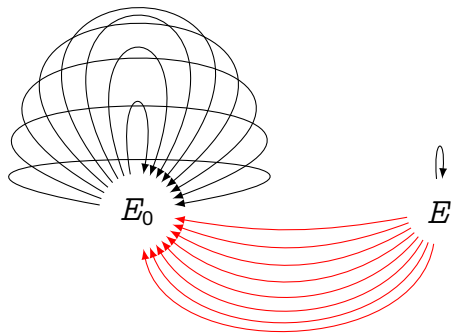
$$\mathcal{O} \simeq \alpha \mathcal{O} \alpha^{-1}$$



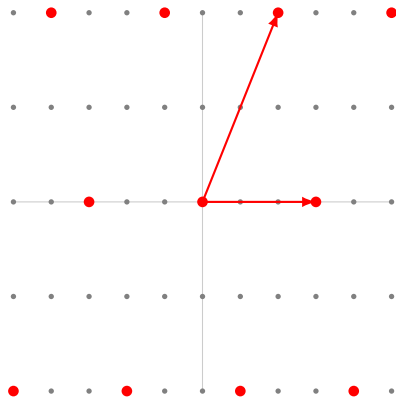
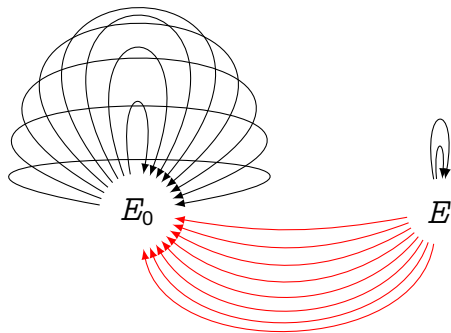
$$\mathbb{Z}[\sqrt{-q}]$$



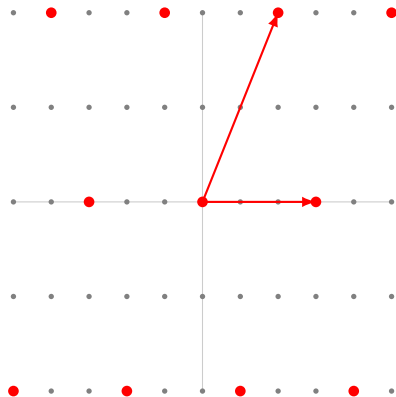
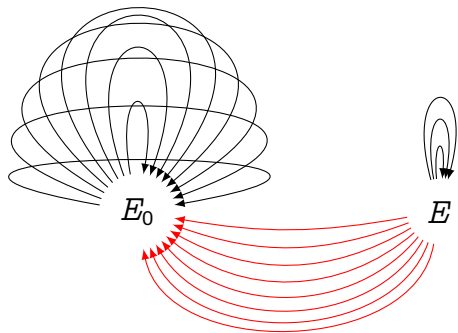
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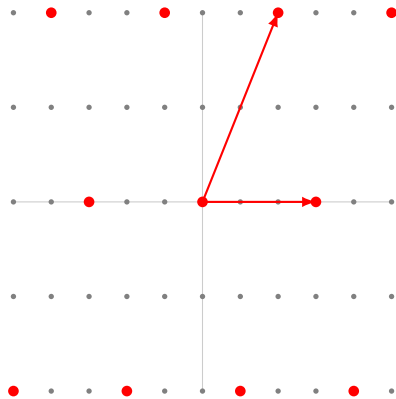
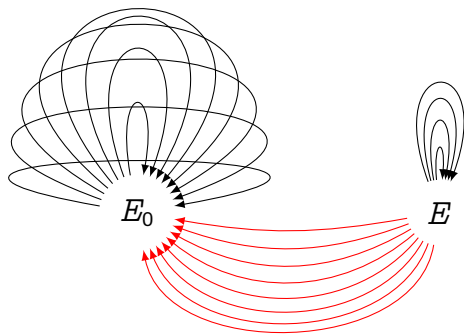
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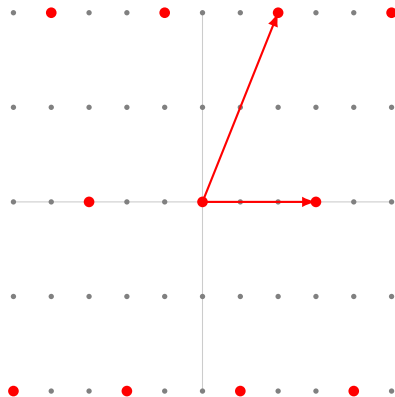
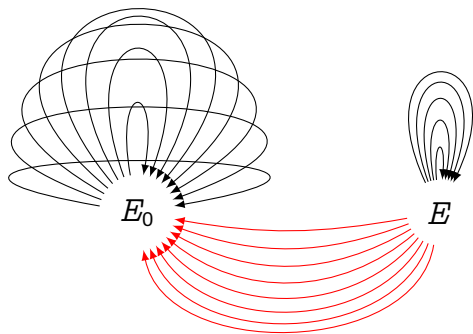
$$\mathbb{Z}[\sqrt{-q}]$$



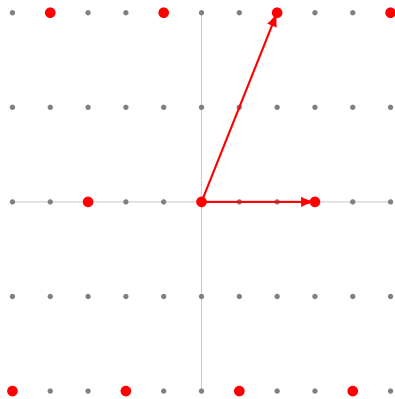
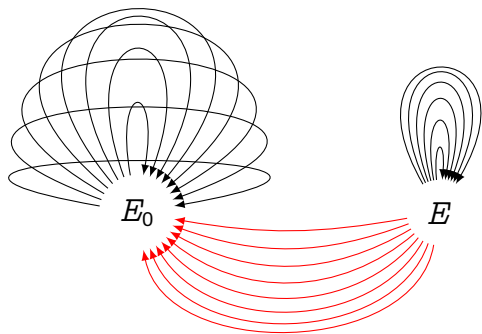
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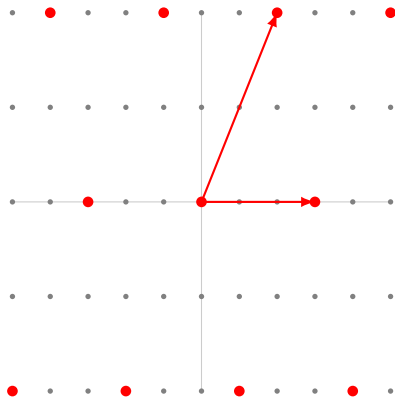
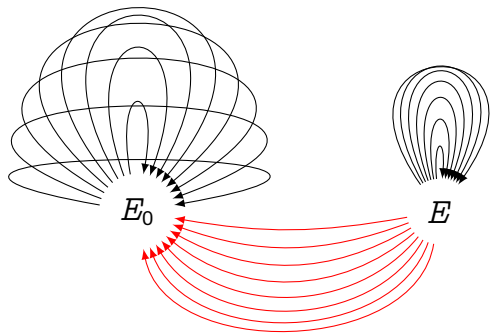
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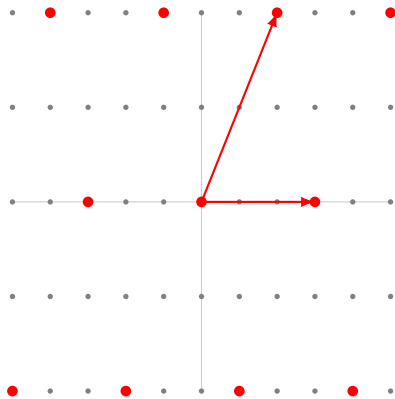
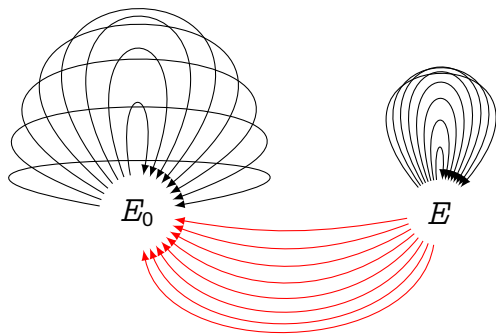
$$\mathbb{Z}[\sqrt{-q}]$$



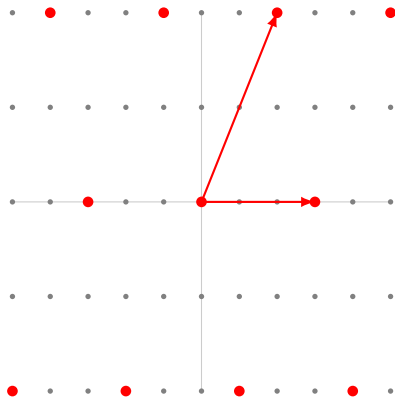
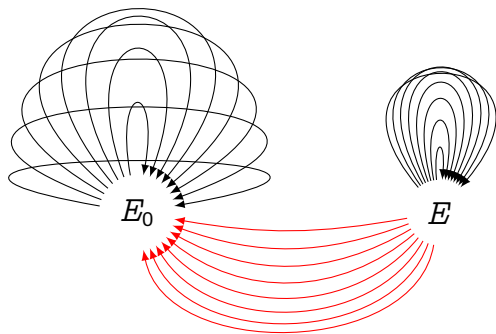
$$\mathbb{Z}[\sqrt{-q}]$$



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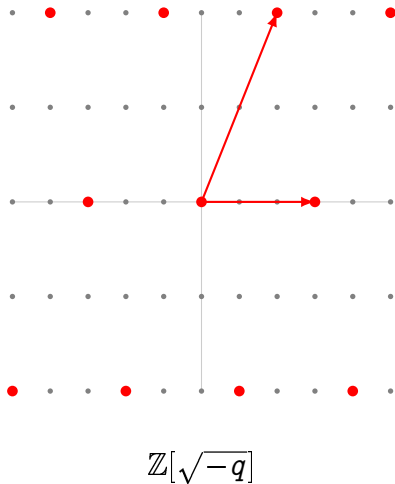
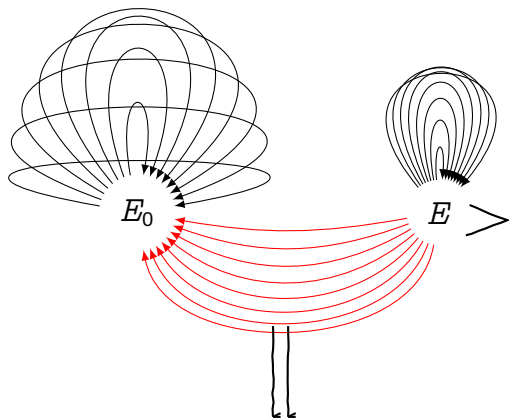


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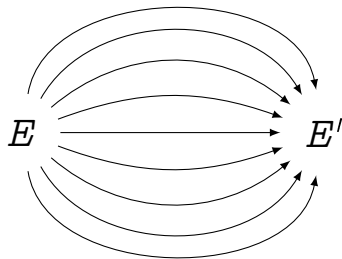
$$\mathbb{Z}[\sqrt{-q}]$$

Deuring: $\text{End}(E_0)$ sublattices $\leftrightarrow$ homsets $\leftrightarrow$ curves isogenous to E_0 .



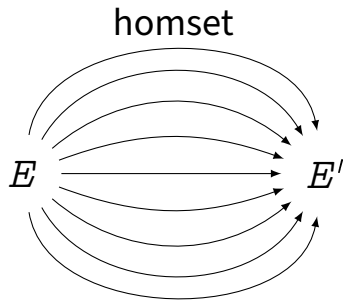
Deuring: $\text{End}(E_0)$ sublattices $\leftrightarrow$ homsets $\leftrightarrow$ curves isogenous to E_0 .

homset



lattice (class)

$$\begin{pmatrix} 13 & 3 & 4 & 1 \\ 0 & 13 & 14 & 0 \\ 0 & 0 & 31 & 9 \\ 0 & 0 & 0 & 173 \end{pmatrix}$$



degree

lattice (class)

$$\begin{pmatrix} 13 & 3 & 4 & 1 \\ 0 & 13 & 14 & 0 \\ 0 & 0 & 31 & 9 \\ 0 & 0 & 0 & 173 \end{pmatrix}$$

norm

The supersingular space

- $\approx \frac{p}{12}$ curves defined over $\mathbb{F}_{p^2}$
- of which $\sim \sqrt{p}$ curves defined over $\mathbb{F}_p$
- Connected by ℓ -isogenies for any choice of $\ell > 1$

The endomorphism ring problem

Given a random
supersingular curve E ,
compute a basis of $\text{End}(E)$.



Given random supersingular
curves E, E' , compute an
isogeny $E \rightarrow E'$.

The endomorphism ring problem

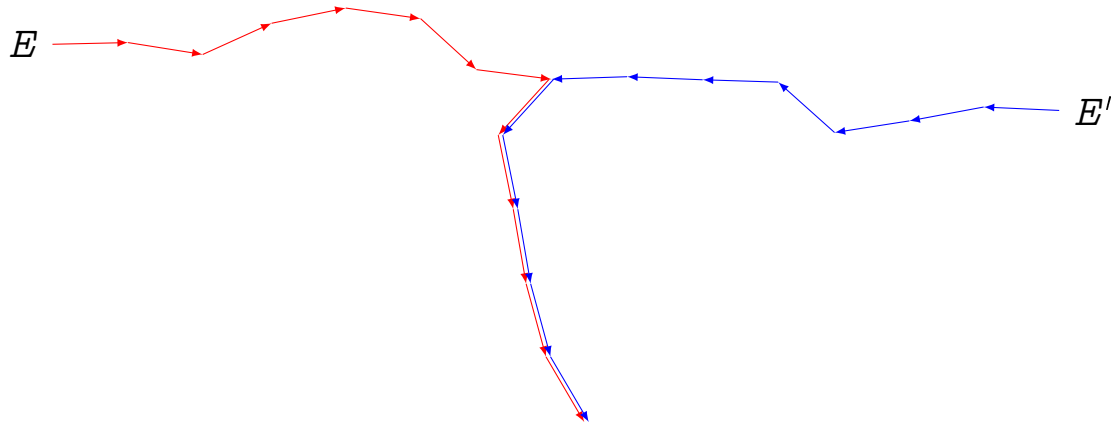
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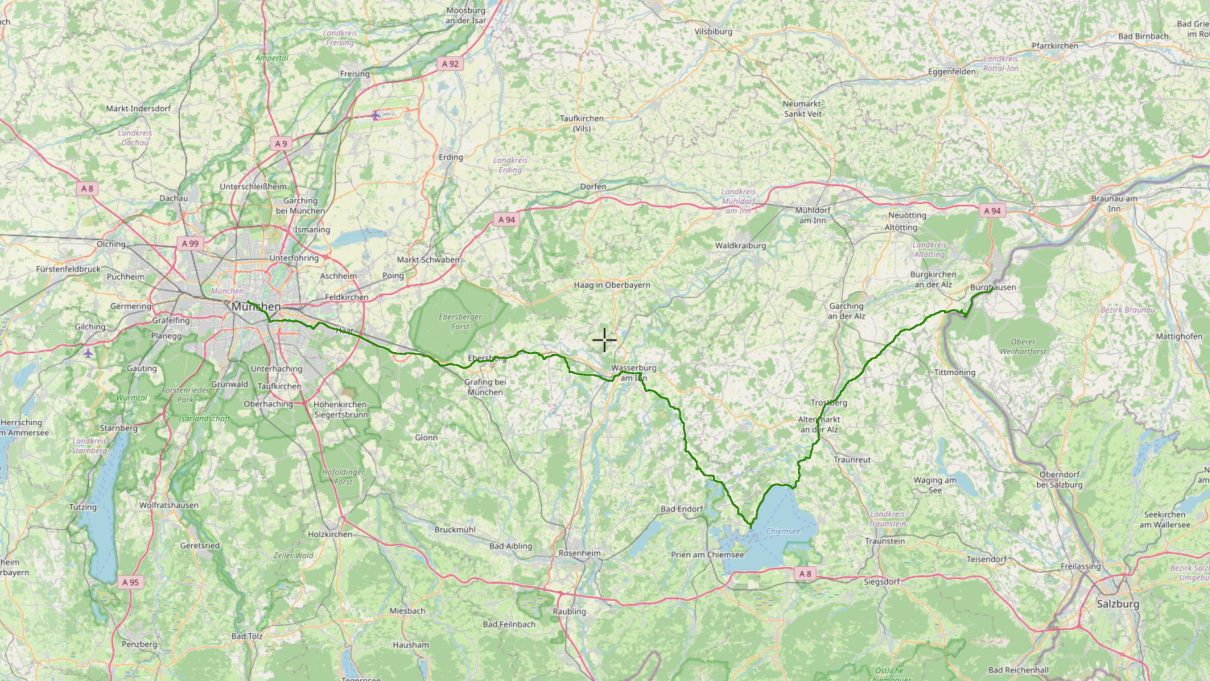


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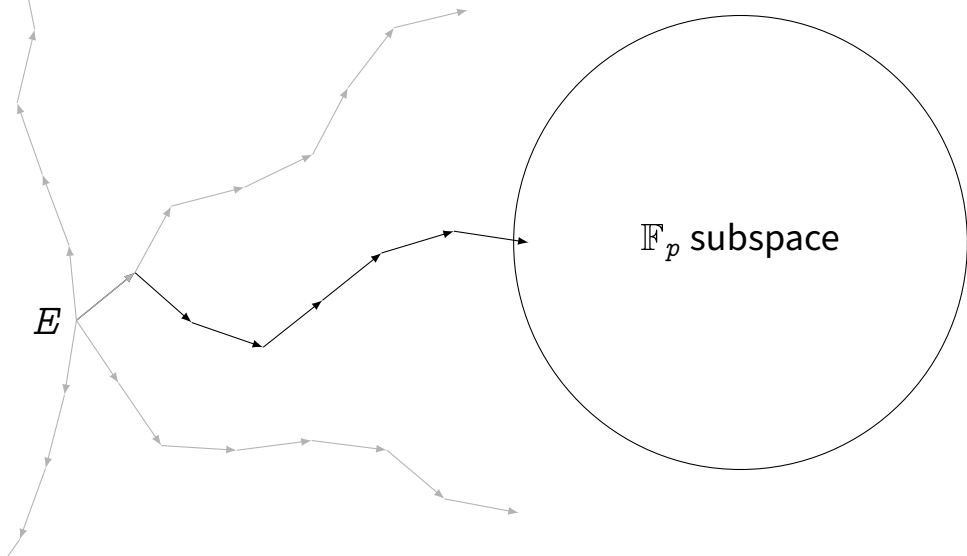
Both problems are random self-reducible

Collision search: Galbraith '99





Collision search: Delfs-Galbraith '16



Frobenius

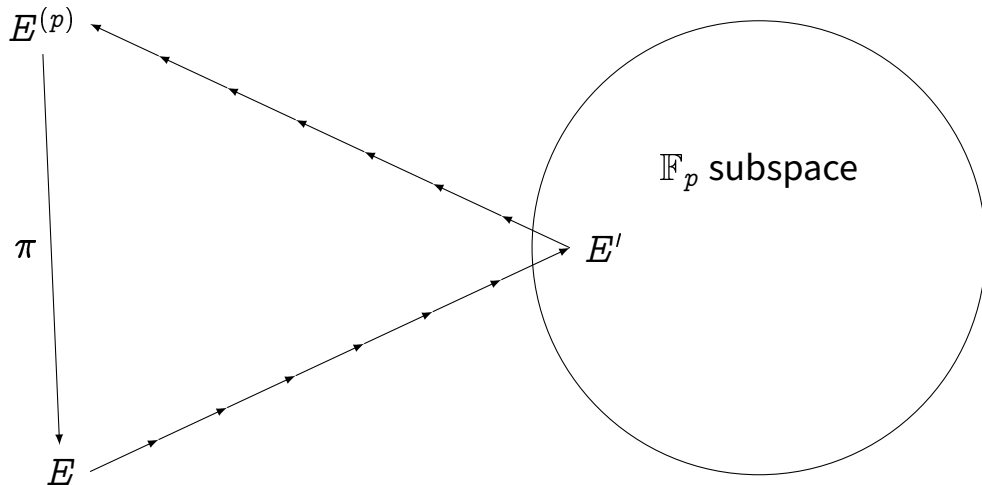
$$E : y^2 = x^3 + ax + b$$

$$\pi : E \longrightarrow E^{(p)}$$

$$(x, y) \longmapsto (x^p, y^p)$$

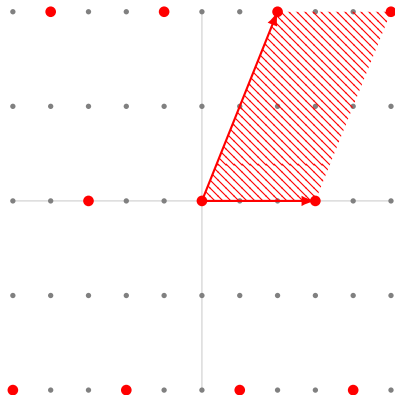
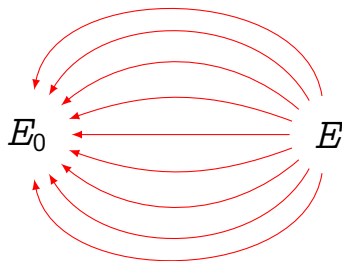
$$E^{(p)} : y^2 = x^3 + a^p x + b^p$$

Reflections (Chenu-Smith '21)



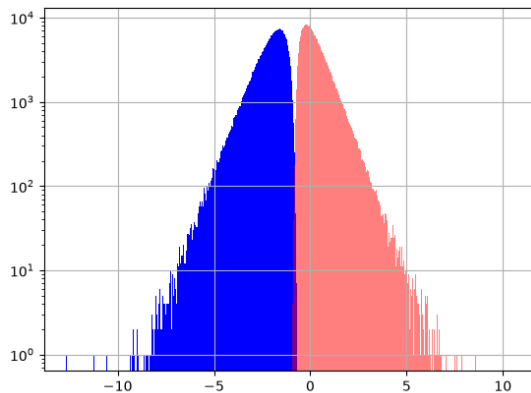
Geometry of numbers

discriminant = covolume = p

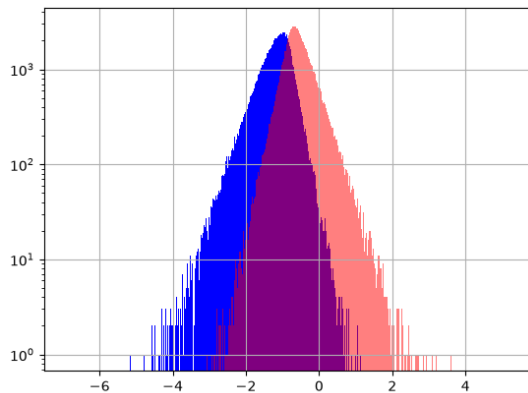


Gaussian heuristic $\Rightarrow d_1 \approx d_2 \approx d_3 \approx d_4 \approx \sqrt{p}$

Squared minima of random $\text{Hom}(E, E')$, log scale



1st and 4th minimum



2nd and 3rd minimum

How hard is it to find a short isogeny? (Folklore)

$$E \xrightarrow{\deg \varphi \approx \sqrt{p}} E'$$

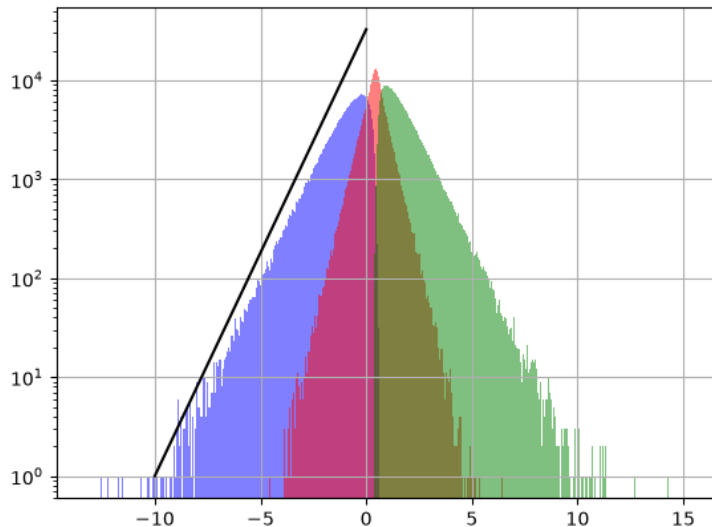
- $O(n)$ isogenies of degree n from E
- $\Rightarrow O(n^2)$ isogenies of degree $\leq n$ from E

How hard is it to find a short isogeny? (Folklore)

$$E \xrightarrow{p^{1/4}} \longleftrightarrow \xleftarrow{p^{1/4}} E'$$

- $O(n)$ isogenies of degree n from E
- $\Rightarrow O(n^2)$ isogenies of degree $\leq n$ from E
- $\Rightarrow$ If we're lucky, we can meet-in-the-middle at cost $O(\sqrt{p})$.

Minima of endomorphism rings



Duality

$$\begin{array}{ccc} & \mathrm{Hom}(E_1^{(p)}, E_2) & \\ \swarrow & & \swarrow \\ \mathrm{Hom}(E_1, E_2) & \# & \mathrm{Hom}(E_1^{(p)}, E_2^{(p)}) \\ \searrow & & \searrow \\ & \mathrm{Hom}(E_1, E_2^{(p)}) & \end{array}$$

Transference

$$\mathrm{Hom}(E_1, E_2)$$

$$d_1, d_2, d_3, d_4$$

$$\mathrm{Hom}(E_1, E_2^{(p)})$$

$$d_1^*, d_2^*, d_3^*, d_4^*$$

$$d_1 d_4^* \approx d_2 d_3^* \approx d_3 d_2^* \approx d_4 d_1^* \approx p$$

Transference

$$\mathrm{Hom}(E_1, E_2)$$

$$d_1, d_2, d_3, d_4$$

$$\mathrm{Hom}(E_1, E_2^{(p)})$$

$$d_1^*, d_2^*, d_3^*, d_4^*$$

$$d_1 d_4^* \approx d_2 d_3^* \approx d_3 d_2^* \approx d_4 d_1^* \approx p$$

$$\mathrm{End}(E)$$

$$1, d_1, d_2, d_3$$

$$\mathrm{Hom}(E, E^{(p)})$$

$$d_1^*, d_2^*, d_3^*, d_4^*$$

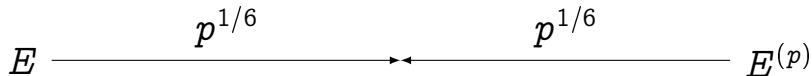
$$d_4^* \approx d_2 d_3^* \approx d_3 d_2^* \approx d_4 d_1^* \approx p$$

Wesolowski's attack '26

$$E \xrightarrow{p^{1/3}} E^{(p)}$$

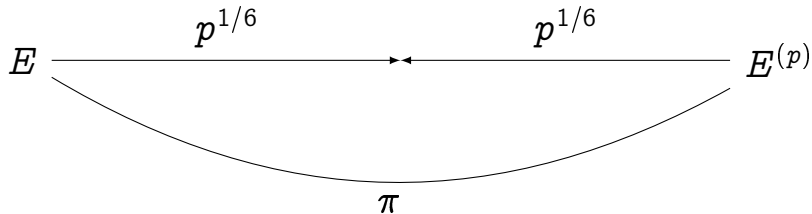
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Wesolowski's attack '26



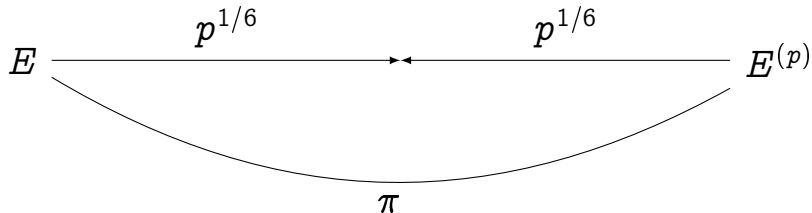
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Wesolowski's attack '26



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Wesolowski's attack '26



- $O(n)$ isogenies of degree n from E
- $\Rightarrow O(n^2)$ isogenies of degree $\leq n$ from E
- $\Rightarrow$ If we're lucky, we can meet-in-the-middle at cost $O(p^{1/3})$ time **and memory!**

How to cost Wesolowski's attack?

Invariant: $T^2 M = \tilde{O}(p)$ via

- van Oorschot–Wiener
- “going under” (Chenu-Smith)

At constant memory: $T = \tilde{O}(\sqrt{p})$

With memory bound w : $T = \tilde{O}(\sqrt{p/w})$

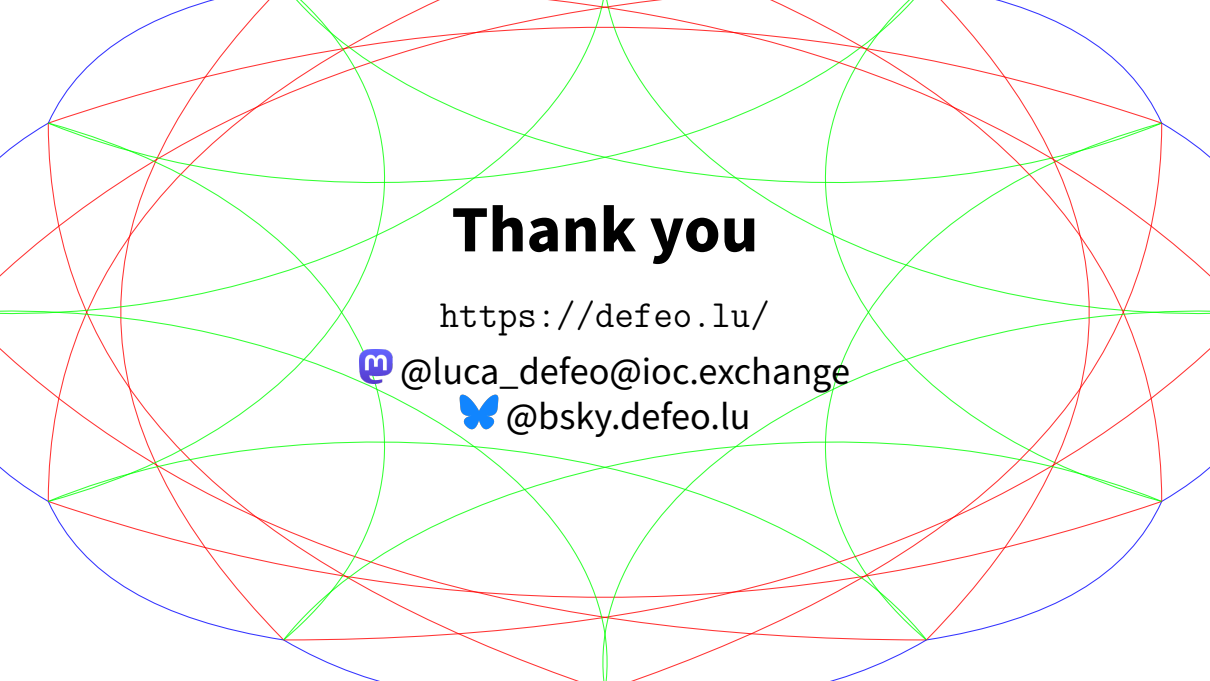
SQIsign Round 3

	old p <i>bits</i>	new p <i>bits</i>	memory <i>bound</i>	$\sqrt{p/w}$ <i>bits</i>	PK <i>bytes</i>	Sig <i>bytes</i>
NIST-I	250.3	325.6	2^{80}	245.6	83	200
NIST-III	382.0	504.8	2^{120}	384.8	129	306
NIST-V	504.8	668.1	2^{160}	508.1	169	406

SQIsign Round 3

	old p <i>bits</i>	new p <i>bits</i>	memory <i>bound</i>	$\sqrt{p/w}$ <i>bits</i>	PK <i>bytes</i>	Sig <i>bytes</i>
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	Key Gen.	Signing <i>Mcycles</i>	Verification
NIST-I	32.2	93.9	12.1
NIST-III	102.1	320.3	31.5
NIST-V	214.5	674.7	77.6



Thank you

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